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Victoria M. Garibay

Professional Summary

I am a computational data scientist and engineer whose experience includes large-scale data integration, metric development, applied machine learning, and delivering production-ready services. I am practiced in Python, R, SQL, full-stack development, cloud-based data pipelines, and cross-functional collaboration.

Education

Aug. 2018-Aug. 2022 | Purdue University | West Lafayette, IN

2022 – Doctor of Philosophy in Agricultural & Biological Engineering

Ecological Sciences & Engineering Interdisciplinary Graduate Program

Dissertation: Water Management Solutions for East Africa: Increasing Availability and Utilization of Data for Decision-Making

May 2011-Dec. 2017 | Texas A&M University | College Station, TX

2017 – Master of Science in Biological & Agricultural Engineering

Thesis: Irrigation Efficiency Study Based on DSSAT CSM-CROPGRO Cotton Model Calibrated with In-Season Data from the Texas High Plains

2015 – Bachelor of Science in Biological & Agricultural Engineering

Environmental and Natural Resource Engineering Emphasis

Experience

Sep. 2025-Present | Freelance/Self-Employed | Remote

Independent AI/ML Engineer, Data Scientist, & Software Developer

- Built a serverless RAG LLM-based chat assistant with Azure Functions and Azure AI Search, implementing a CI/CD pipeline and automated tests for maintainability
- Developed a geospatial scoring pipeline (Postgres, GIS) for an interactive Power BI dashboard to inform accommodation site placement
- Developed a fully online dashboard app (Next.js, React, TypeScript, Redux) with RESTful APIs and a Neon-hosted PostgreSQL backend with automated CI/CD cloud deployment (Vercel)

Sep. 2022-Present | University of Amsterdam | Amsterdam, NH, Netherlands

Postdoctoral Researcher-Informatics Institute

- Studied interactions between climate change adaptation and poverty traps for marginalized communities through development of an agent-based socioeconomic model
- Applied the final simulation tool to identify factors reinforcing poverty trap dynamics and project the effects of policy implementation
- Incorporated machine learning replacement for agent economic decisions and novel GPU model framework for use with Snellius (Dutch National supercomputer) to scale simulations to one million agents

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Guest Researcher-Informatics Institute

- Generalized the DGL-PTM agent-based modeling framework, which involved troubleshooting and resolving bugs in the new implementation, configuring continuous integration for multiple operating systems and architectures, and containerizing the module to increase flexibility

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Aug. 2018-Aug. 2022 | Purdue University | West Lafayette, IN

Research Assistant-Hydrology/Climate Data

- Performed statistical analyses of climate data using R and quantitatively assessed policies and documentation within and beyond East Africa to determine effects on national data availability
- Investigated feasibility of implementing bias correction methods for modeled precipitation data to improve regional representativeness for East Africa
- Assisted in development of course material on agricultural drainage
- Provided support for LASER-PULSE East Africa Water Security project including development of a regional climate dashboard (Shiny) and management practice scenario watershed study

Teaching Assistant-Statistical Modeling

- Instructed lab sessions and collaborated with colleagues to lead weekly homework help sessions

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Mar. 2018-Aug. 2018 | Texas A&M AgriLife Research Extension Service | College Station, TX

Research Associate

- Collected image data and assisted with in-field drone flights
- Processed geotagged images in several spectra using Pix4D Mapper and ENVI to highlight potential water canal leak locations in southern Texas

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Jan. 2017-May 2017 | United Nations Food and Agriculture Organization | Rome, LZ, Italy

Agroecology Intern-Plant Production & Protection Division

- Assisted in management of Agroecology Knowledge Hub database
- Designed and performed analysis of Intended Nationally Determined Contribution documents for over 100 countries to evaluate interest in crop agriculture-related action (e.g., soil conservation, species diversification, pest and disease control, etc.)

Aug. 2015-Dec. 2016 | Texas A&M University | College Station, TX

Teaching Assistant-Hydrology/Waste Management/Air Quality

- Evaluated student performance for fall and spring semesters as well as a six-week study abroad program

Research Assistant-Cotton Quality, Particulate Dispersion Modeling

- Designed and conducted a study on the feasibility of cotton contaminant detection with ultraviolet photography
- Wrote reports, edited journal articles, and processed data from the air quality research lab

Supervisory references and/or contact information available upon request for all positions

Skills & Qualifications

- Practiced in scientific research, technical writing, and programming (Python, R, SQL, MATLAB)
- Skilled in data management, including data handling, metadata and provenance tracking, relational databases (PostgreSQL, Access), and data cleaning
- Experienced in GIS (QGIS, ArcGIS), data and visualization libraries/software (pandas, numpy, dplyr, matplotlib, seaborn, ggplot, Tableau, Origin, etc.), graphics (Autodesk, Adobe Photoshop, GIMP), and various office application suites
- Comfortable with cluster computing, cloud computing (Azure), software development best practices, model evaluation, metric development, and select supervised, unsupervised, and reinforcement machine learning techniques
- Experienced in developing and deploying interactive dashboards and visulhalizations using web technologies (TypeScript, React, Next.js, Redux), Power BI, and Shiny
- Proven ability to coordinate interdisciplinary technical projects and communicate findings to technical and nontechnical audiences
- Certificate in Foundations of College Teaching
- Intermediate Spanish Level B1 Certificate
- Quantum Programming Workshop for Purdue Engineers and Scientists Certificate

Peer -Reviewed Publications

- **V.M. Garibay** & D. Roy. "Pricing Adaptation: A Simulated Study of Poverty Intervention Strategies Under Climate Shocks" Journal of Computational Science [2026, Accepted]
- **V.M. Garibay.** "Accelerated Approximation of Bellman Equation Solutions: Agent Policy Optimization With a Feedforward Neural Network" Lecture Notes in Computer Science (2025):15908.
- DOI: 10.1007/978-3-031-97557-8_17
- V. Bazyleva, **V.M. Garibay**, & D. Roy. "Trajectory-based global sensitivity analysis in multiscale models" (2024) Scientific Reports 14: 13902.
  DOI: 10.1038/s41598-024-64331-x
- V. Bazyleva, **V.M. Garibay**, & D. Roy. "Global Sensitivity Analysis Using Polynomial Chaos Expansion on the Grassmann Manifold" (2023) Lecture Notes in Computer Science: 14077.
  DOI: 10.1007/978-3-031-36030-5_46
- **V.M. Garibay**, M.W. Gitau, V. Kongo, J. Kisekka, & D. Moriasi. "A Comparative Evaluation of Water Resource Policy Inventories Towards the Improvement of East African Climate and Water Data Infrastructure" Water Resources Management 36 (2022): 4019–4038.
  DOI: 10.1007/s11269-022-03231-z
- **V.M. Garibay**, M.W. Gitau, N. Kiggundu, & D. Moriasi. "Evaluation of Reanalysis Precipitation Data and Potential Bias Correction Methods for Use in Data-Scarce Areas." Water Resources Management 35 (2021): 1587–1602.
  DOI: 10.1007/s11269-021-02804-8
- **V.M. Garibay**, K. Kothari, S. Ale, D.C. Gitz, G. Morgan, & C. Munster. "Determining water-use-efficient irrigation strategies for cotton using the DSSAT CSM CROPGRO-cotton model evaluated with in-season data." Agricultural Water Management 223 (2019): 105695.
  DOI: 10.1016/j.agwat.2019.105695